EE660: Mathematical Foundations and Methods

Fall 2026

Date: 8/25/2026

Important: EE660 materials

- **There is no official textbook!**
- **Instead, we will use the course lecture notes:**
  - https://stephentu.github.io/pdfs/EE660_Lecture_Notes.pdf
  - These will be updated frequently, make sure to refresh!
- **Additionally, the course syllabus is available here:**
  - https://stephentu.github.io/pdfs/EE660_Fa2025_Syllabus.pdf
- **All logistic information covered today is also covered in the syllabus.**

EE660 staff

- **Prof. Stephen Tu**
    - OH: Thursday 3-4pm, EEB 326 **(starting next week)**
- **TA: Yichen Zhou**
    - OH: Friday 3-4pm, PHE 410 **(starting next week)**

EE660 discussion section

- EE660 discussion section is held on **Friday, 2-2:50pm, OHE 100D**
    - TA will review material, go over example problems, answer questions.
- **We will start this Friday (8/28)!**

EE660 grade breakdown

Assignment	Weight
In-Class Quizzes	30%
Midterm	30%
Final	40%

- **Midterm:** 10/7 during class time, **in class.**
- **Final:** 12/10, 4:30pm-6:30pm (Location TBD).
  - Please double check https://arr.usc.edu/final-exam-schedule/#chapter=fall-2026 and make sure I computed the final exam time-slot correctly!!!
- **Note:** we **cannot** make any exceptions to the dates of in-class quizzes, midterm, and final.
  - If you cannot make any of these times, then you cannot take this class!

EE660 homework

- There will be 5 **optional** homework assignments.
  - You do **not** need to turn them in.

EE660 in-class quizzes

- There will be 5 **in-class quizzes**, which will run from **4:05-4:25pm** at the beginning of lecture.
- These quizzes are **closed-book**, you may not use any notes.
- Each quiz will ask exactly one question **taken from the homework.**

Things we will not have

- Attendance points.
- Class participation points.
- Class project.

Any logistical questions?

EE660 Course Content

- EE660 focuses on the mathematical **foundations** and **algorithms** behind supervised and unsupervised learning.

- We will use mathematical tools from multivariate calculus, linear algebra, and probability theory to **analyze** when/why algorithms work.

    - We will write mathematical proofs in class and in quizzes/exams.

- As a result, the real prerequisite for this course is both **comfort and interest** in mathematical arguments.

EE660 Course Content

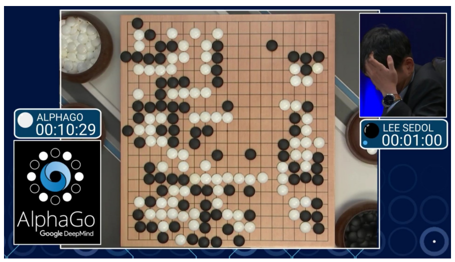
- We will focus a lot on “classical” machine learning:
  - **Supervised:** Perceptron, SVM, kernels, empirical risk minimization, PAC learning, Rademacher complexity, VC-dimension, etc.
  - **Unsupervised:** random projections, PCA, maximum likelihood estimation, expectation-maximization, etc.
- We will also study more “modern” machine learning:
  - Overparametrization in neural networks, variational inference, sequence modelling, recurrent neural networks.

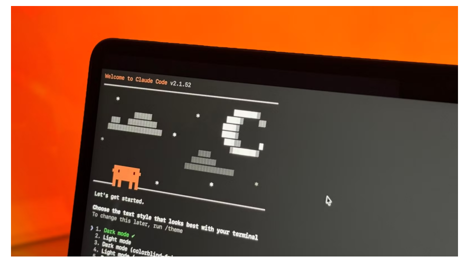
EE660 Course Content

- We will **not** cover the latest deep learning / LLM architectures, such as:
    - Should I use ReLU, swish, GeLU, SwiGLU, or some other activation?
    - Should I use LayerNorm, RMSNorm, BatchNorm, etc?
    - Should I use a constant learning rate, warm-up + cosine decay schedule, random restarts, etc?
    - Should I use train in FP32, BF16, FP16, etc?
- For more hands on experience with deep learning: **EE 641.**

Why do we care about mathematical foundations?

- Machine learning works amazingly well…





- Until it does not…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How OpenAI Lost Control of an AI Model—and What Needs to Change

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by Harry Booth
BOOTH IS A REPORTER AT TIME.

JUL 24, 2026 8:13 AM PT

OpenAI CEO Sam Altman talks to reporters at the Dirksen Senate Office building in Washington, DC on June 3, 2026. *Nathan Posner—Anadolu/Getty Images*

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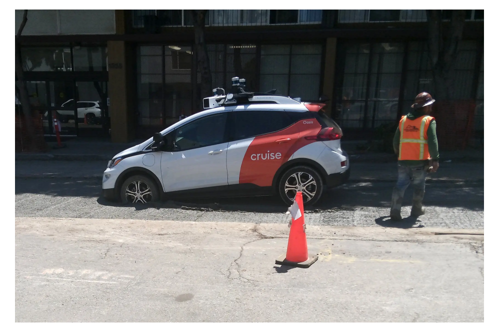
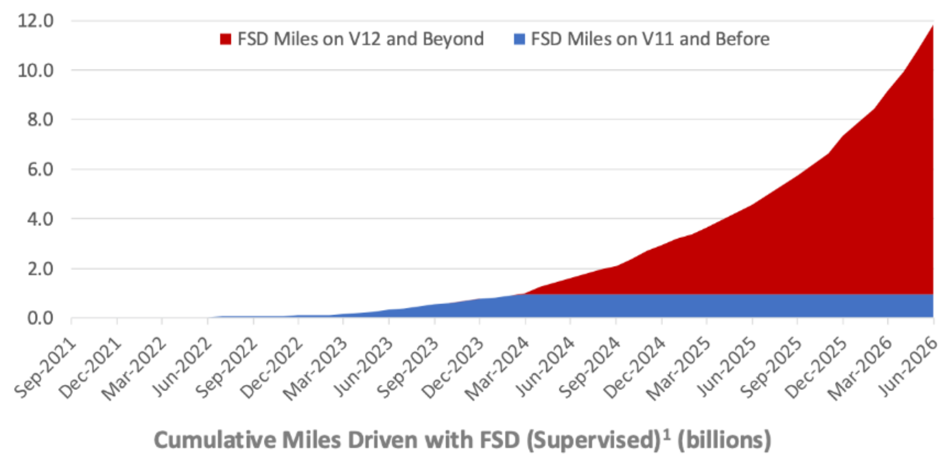


Limits of large-scale evaluation

- Absent any formal models, the only way to evaluate a system is through large-scale experimentation.

- This of course has its own drawbacks:

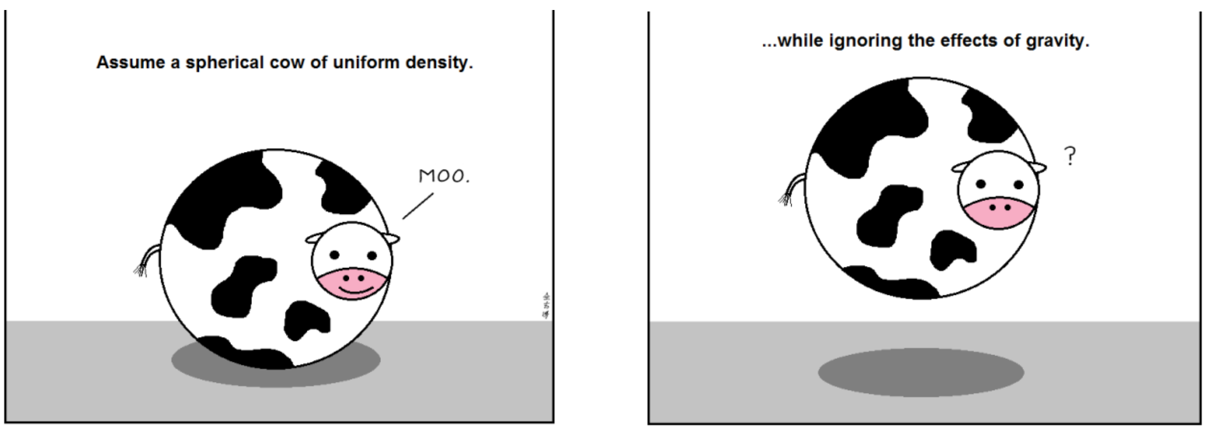
  - **Ex:** Tesla FSD has logged over **9B driving miles**, but is still not fully autonomous (requires human intervention).



Tesla Q2 2026 Earnings Report

The need for rigorous understanding

- We take inspiration from science/engineering disciplines.
- Develop simple mathematical models where we can understand everything as much as possible.
- Use these mental frameworks to design complex systems at scale.



Simple mathematical models -> real-world impact

- In past ~5 years, quite a few success stories in the LLM-space arise from considering simple mathematical models:

  - **GPTQ / QuIP** (quantization for LLMs) uses many ideas from randomized linear algebra, concentration of measure.

  - Direct preference optimization (**DPO**) uses an exact solution to KL-regularized optimization to improve RHLF.

  - **LoRA** uses low-rank matrices for fine-tuning large-scale foundation models.

Part I: Supervised learning

- **Generalization:** What does good prediction on training imply about prediction on test?

- **PAC learning:** What are the limits of learning? When is learning even possible? When is it not?

- **Algorithms:** What kind of algorithms should we use to learn?

Part II: Unsupervised learning

- **Dimensionality reduction/clustering:** What techniques can we use to reduce dimensionality and cluster data, when we do not have labels available?

- **Latent variable models:** How do we do inference where there hidden latent structure in our observations?

- **Sequence modeling:** How do we learn from and make predictions when data is sequential.

ChatGPT/Claude/Gemini

- You are free to use these AI tools to help in your studies:
    - Explaining concepts in different ways, including via concrete examples and visualizations.
    - Coming up with practice exam questions.
    - Evaluating your understanding of a concept.
- **But of course:**
    - *****None of these tools will be available on the in-class quizzes/ exams.*****

Any questions?